

Detection & Classification of Spoofing Attacks on GNSS Signals Using AI Models on NavIC System

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Abstract. Global Navigation Satellite Systems (GNSS) have become an indispensable part of our contemporary infrastructure for navigation and timing, being employed in several mission critical applications ranging from unmanned aerial vehicles (UAVs), autonomous transportation systems to aviation, maritime navigation and communication networks. Nevertheless, even when there exists such a widespread use, GNSS is still afflicted by weak authenticity of transmitted signals (spoofing); that is an attacker could send counterfeit signals, which can deceive the receiver involved in calculating wrong position or velocity or timing information. These attacks represent a great risk especially in safety critical contexts where the non-detection of even an only spoofing is enough to cause catastrophic events.

In this paper, we propose an integrated artificial intelligence-based framework for GNSS spoofing attack detection on real signal data. Three sophisticated machine learning models like Deep Neural Networks (DNN), Random Forest (RF) and Extreme Gradient Boosting (XGBoost) were constructed, trained and tested to see how well they could distinguish genuine and spoofed signals. The achieved cross-sections are based on realistic data sets.

Keywords: GNSS security, Spoofing Attacks, Positioning Navigation and Timing (PNT), Machine learning, Extreme Gradient Boosting.

1 Introduction

Global Navigation Satellite Systems (GNSS) have become critical to the functioning of our modern technology-based societies through Positioning, Navigation and Timing (PNT). These services drive civilian and military applications ranging from smartphones navigation and fleet management to aviation essential systems, maritime navigation, as well as controlling autonomous unmanned platforms. Because critical and safety-relevant applications are increasingly based on GNSS there is a growing need to achieve integrity and reliability of satellite-based navigation. Finally, GNSS signals, while being a great technology to have offer very low-powered signals once they reach ground level and are susceptible. Spoofing attacks are especially harmful,

since such attacks do not typically bring down the service, but instead lead the receiver to accept incorrect yet believable navigation information. Thus, the attacked entity may function properly relying on invalid positional or temporal information and therefore, such kinds of attacks are specifically challenging to discover. Spoofing detection via Signal Processing: The diagnostic techniques for detecting GNSS spoofing have remained largely unchanged over the years and have included a blended use of signal processing (e.g., signal power, angle-of-arrival, cryptographic authentication, and multi-sensor consistency checks). Although successful in some cases, they typically involve costly hardware, multiple antennas, or knowledge of attack profile. What is more, their accuracy may weaken in complex environments with multipath propagation, dynamic motion, or advanced attacks that closely replicate the behaviors of legitimate signals [5].

The rapid development of artificial intelligence and machine learning capabilities hold new potential to develop GNSS spoofing mitigation approaches that are able to adapt and learn from patterns and experiences [6]. In general, data-driven models can learn intricate and nonlinear relationships between signal features that may not be easily captured by rule-based methods. Machine learning models accomplish this by learning earlier signal patterns and identifying the minutest of variations to statistical characteristics caused by spoofing even when counterfeit signals have almost similar characteristics to authentic signals.

In light, this paper aims to explore the use of complex machine learning models (DNN, RF and XGBoost) in GNSS spoof detection from a practical standpoint, using real world data. Special attention is paid to reducing false-negative mistakes taking into consideration their negative and immediate impact on navigation safety. Based on systematic comparative analysis and real-time [7],[8], [9].

Proposed Objectives

This research is oriented towards building, developing and testing as an Artificial intelligence-based framework of the essential points necessary for the operational reliability in applied real-life experiences and in real time regarding the performance of spoofing attacks on Global Navigation Satellite System (GNSS) applications. This work aims to overcome the classic spoofing detection approach limitation by using advanced learning machine models, which can learn more complex signal behavior in nominal and spoofed environments. The motivation behind this work is the recent growth of GNSS dependency in safety-critical applications such as unmanned aerial vehicles, autonomous navigation systems and timing-critical infrastructures, where even a single undetected spoofing attack can lead to catastrophic operational and security consequences. Therefore, the goal is not just to maximize classification accuracy, but also to minimize false-negative errors for reliable and timely detection of spoofing attacks across varying environmental conditions. In order to fulfil this aim, the following specific objectives are set:

Goal 1: Building a Non-ideal GNSS Spoofing Detection Dataset.

The last of these three is more concerned with authentic and spoofer GNSS signal data usage and processing [1]. RAW data omit environmental noise, multipath, receiver imperfections, and time-varying signals, which makes the proposed detection framework characterized close to real operation condition [2].

Goal2: Engineering features and characterizing the signal

Its goal is to obtaining and creating informative features from unique view GNSS signals measurements. Essential factors are the C/N_0 (carrier-to-noise density ratio), Doppler shift, pseudo range error, and position changes. A number of extra features including motion statistics, signal stability indicators and local variation statistics are computed to capture fine spoofing patterns unable to be learned by the models. Design and Development of Multiple AI Models.

Goal.3: Model development and investigation

To achieve this goal, several machine learning models can be designed and implemented, specifically Deep Neural Networks (DNN), -to investigate the merits and demerits of each method within GNSS spoofing detection and to perform a comparative analysis across learning architectures.

Goal. 4: Comparison + Error Analysis

The goal of this task is the systematic comparison of model performance with standard metrics such as accuracy, recall, precision, and the F1-score needed for confusion matrices. The analysis of false-negative errors is focused upon because any such errors directly affect navigation system safety.

Methodology

The complete methodological approach used in this research to detect spoofing attacks on Global Navigation Satellite System (GNSS) signals using artificial intelligence and machine learning. The methodology is structured to ensure that the results can be replicated, that different models can be fairly compared, and that the approach can be applied effectively in real-world settings. It covers all the key stages, including data collection, preprocessing, feature engineering, model development, training strategies, and methods for evaluating performance.

The research employs real-time GNSS signal datasets that are comprised of real as well as spoofed GNSS signal's due to it being very important that only real GNSS signal datasets, as in, the signal datasets from real-life environments are utilized because they reflect the actual conditions under which the GNSS signals propagate, namely, thermal noise, multipath reflection of GNSS signals, atmospheric disturbances to the GNSS signals and inaccuracies related to GNSS hardware/receivers, all of which cannot be accurately represented with synthetic GNSS signal datasets. These GNSS Signal Datasets were collected from several publicly available data repositories that focus on

discovering an analyzing GNSS jamming and spoofing capabilities, as well as from Operational GNSS user receivers (i.e., GNSS Support Systems, such as NavIC; GPS, etc.). Each data sample was assigned a binary classification of Authentic or Spoofed, allowing for the supervised technique of learning to be applied [3], [4].

The datasets include multiple GNSS signal attributes including, but not limited to;

- i. C/N_0
- ii. Doppler frequency shift
- iii. Pseudo range error
- iv. Geographic Position estimations (East-North-Height)
- v. Temporal Parameters of GNSS Signal Stability.

To balance class distributions to decrease model bias, the dataset was split into Training, Validation, and Test set components for training and evaluation.

Data Preprocessing and Feature Engineering

Data preprocessing is an important process that must occur before model training occurs in order to increase the quality of the data used in learning and to facilitate learning efficiency. The following steps were performed on the datasets during preprocessing: Removal of all missing or corrupted samples from the datasets; and the Identification of all outliers from the data by employing standard statistical methods to identify and remove all outliers due to measurement errors (e.g., sudden jumps) or receiver malfunction (e.g., "flickering" RF signal).

To enhance the ability of each model to detect the presence of spoofing, Feature Engineering also played a large role in the overall effectiveness of the final model outputs. Several features were generated to capture Signals within the time domain (i.e., temporal behavior) and the degree of Stability, in addition to the typical raw signal values (C/N_0).

The following features were generated from the feature engineering experiments:

1. Differences and Ratios between C/N_0 values. This feature assists in detecting a sudden drop-off of signal strength or an increase in power level.
2. The Moving Mean and Standard Deviation of both C/N_0 and Doppler Shift measurements over a fixed-length time window.
3. Metrics for local variation. These metrics were designed to detect large and rapid changes in Position.
4. Indicators for Temporal Consistency (including drift and pull-in) for APTs (Advanced Planning Technologies).

Machine Learning Models and Implementation

To determine which of three different machine learning models would be the most effective at predicting GNSS spoofing events, three separate machine learning models were developed and assessed for performance.

Once all numerical features were generated and identified, they were scaled using standard Scaling methods to provide balanced contributions from all features. The normalization process prevents features with large numerical ranges from dominating other features.

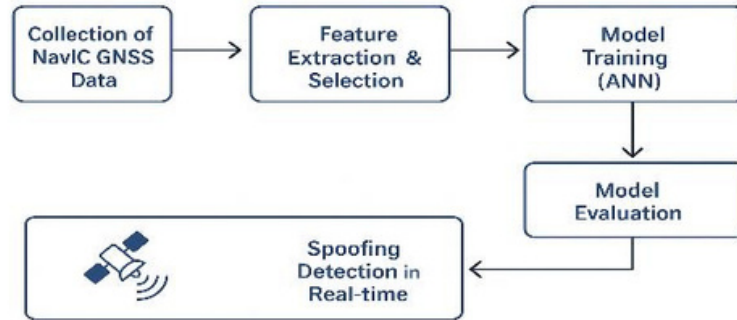


Fig. 1. ProposedSystem Workflow

Training Strategy & Metrics

For comparison purposes, all the Models were developed under the same conditions. The dataset was separated such that 80% of the data was used for training and 20% was used for testing model performance, together with a separate validation set that was used to assess the ability of the Models to generalise to new data.

Experimental Workflow

The workflow of this experimental study starts with the acquisition of raw GNSS signals from GNSS receivers, continues with the data preprocessing stage and feature extraction stage, and ends with feeding the extracted feature vectors into Machine Learning Models for both training and validating. Finally, the output of the different Models will be evaluated and compared with the objective of identifying the Model that is most reliable in detecting Spoofing. The experimental workflow of this study is illustrated in a simple flowchart, which details the processing pipeline from the time signals are received until a final decision has been reached and all Mitigation and Response Mechanisms are ready.

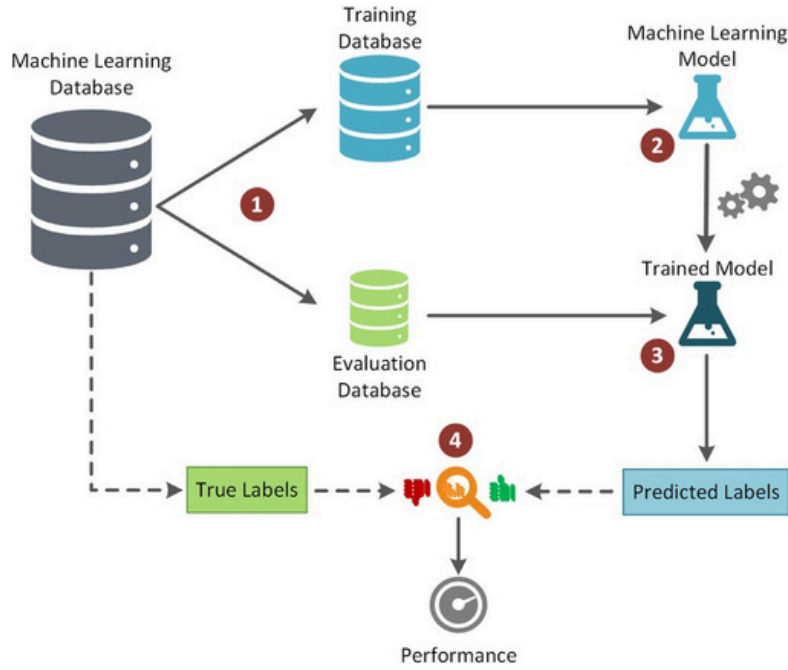


Fig. 2. Proposed Model working process

Results & Discussion

the experimental results derived from the use of artificial intelligence and machine learning models to detect GNSS spoofing attacks. This chapter also offers a comprehensive and critical evaluation of the model performance based on accuracy, reliability, computational efficiency, and their performance when utilized in real-world operational settings. It is the connection between the Methodological Framework and the Practical Results of this study, as shown in table 1.

Table 1. Result Analysis

Macine Learning Model	Accuracy	Precision	Recall	F1-Score	False Negatives
DNN	94.1	94.6	93.7	94.1	417
Random Forest	93.5	92.8	93.1	92.9	502
XGBoost	95.5	96	95.4	95.7	334

Conclusion

Artificial Intelligence (AI) and Machine Learning (ML) have proven to be essential tools for identifying GNSS spoofing; they can be applied in actual situations where hybrid datasets produce a more accurate result than using conventional simulation techniques. Before developing their proposed research methodologies, the researchers produced a detailed plan outlining how they would implement it. Their plan consisted of: (i) identifying what data they would need and how they would collect it; (ii) using a method that used characteristics of GNSS Spoof and/or MLI in the feature engineering process; (iii) making train datasets for all the models tested (1-1-DNN, RF, and XGBoost), evaluating performance for each, and (iv) conducting a comparative analysis to compare results between models "apples to apples." The results of the experiment demonstrated that XGBoost had the most success overall, classifying the true GNSS Signal from the Spoofed GNSS Signal with an average classification rate above 93% across all three methods. These findings confirm that XGBoost is the most successful model in classifying GNSS Signals; it has the fewest false negatives, the highest recall rate (for classifying true GNSS Signals versus spoofed GNSS Signals), and the highest degree of classifying GNSS Signals for high-risk GNSS Navigation Systems. We found in this study that the integration of artificial intelligence (AI) into Global Navigation Satellite System (GNSS) based systems will serve to enhance both Navigation Security and Reliable GNSS Navigation in today's environment. The outcomes of this study also illustrate that the assessment of Spoofing Detection Systems for Safety Critical Applications must include a more comprehensive set of evaluation metrics than simply accuracy. This is because metrics such as recall and reducing the number of false negatives offer a more realistic means to evaluate Spoofing Detection Systems in Safety Critical Applications. Furthermore, this study has provided a robust scientific and practical basis for the construction of smart and Real-Time Deployable GNSS Spoofing Detection Systems. Therefore, the results of this study will have tremendous benefits for the Satellite Navigation Security field. The next step is to further research the integration of this work into real-world Application Platforms such as: Unmanned Aerial Vehicles, Autonomous Vehicles, and Timing- and Location-Critical Infrastructure.

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